Venkata Lakshmi Parimala Pasupuleti

AI/ML Engineer & Senior Software Engineer · Python / LangGraph / AWS · LLMs & Generative AI · MLOps · Data Pipelines · Full-Stack

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SUMMARY

AI/ML Engineer and Senior Software Engineer with 9 years of production software ownership across AI systems, enterprise data pipelines, compliance-critical backends, and full-stack delivery. Built AgentProbe -- an open-source LLMOps and evaluation platform with CI-integrated eval gate, LLM-as-judge scoring, hallucination detection, and multi-provider model routing. Built LeaseI -- an 8-agent LangGraph system with RAG, fine-tuning workflows, prompt engineering, and HITL guardrails on regulated decisions, evaluated across 1,240 test cases at 93.6% precision. Fine-tuned a custom classification architecture using HuggingFace Transformers and PyTorch; deployed inference API on Quart. Seven years owning compliance-critical Python systems at Entrata (100K+ MAU, HIPAA/PHI, HUD, AWS) with zero violations across the ownership period. Cloud experience across AWS (S3, EC2, RDS, Glue, Lambda), GCP, and Azure. Full-stack delivery in Python, JavaScript/TypeScript, React, FastAPI, PostgreSQL, RabbitMQ. M.S. Data Analytics Engineering, George Mason University (GPA 3.87, 2026).

TECHNICAL SKILLS

Languages: Python, JavaScript, TypeScript, Java, SQL, PL/SQL, PHP, R, Bash, HTML/CSS, jQuery, AJAX

AI/ML & LLM Engineering: LangGraph, LangChain, Agentic AI, Multi-Agent Orchestration, Large Language Models (LLMs), Generative AI, Fine-tuning, HuggingFace Transformers, PyTorch, TensorFlow, scikit-learn, XGBoost, RAG Pipelines, Vector Databases, ChromaDB, Embeddings, Prompt Engineering, Few-Shot Prompting, Chain-of-Thought Prompting, Tool Calling, HITL, Guardrails, Output Validation, Hallucination Detection, Structured Outputs, LLM Evaluation, LLM-as-Judge, LLMOps, Agent Observability, Agent Tracing, Confidence Scoring, Context Engineering, Pass@k Evaluation, Regression Gate, NLP, BERT, spaCy, OCR, Document Intelligence, OpenAI API, Anthropic API, Groq, Ollama, MCP

ML & Data Science: Random Forest, Logistic Regression, Isolation Forest, Anomaly Detection, Feature Engineering, Statistical Modeling, Model Deployment, Model Monitoring, MLOps, Anomaly Detection, NumPy, Pandas, Matplotlib, Seaborn, Plotly, ggplot2, Jupyter, Tableau, Power BI, Apache Spark, PySpark, ETL, Data Modeling, Data Pipelines, Data Visualization

Cloud & Infrastructure: AWS (S3, EC2, RDS, Glue, Lambda, Bedrock, SageMaker), Google Cloud Platform (GCP), Azure (App Service, SQL), Docker, Kubernetes (K8s), CI/CD (GitHub Actions), FastAPI, REST APIs, Flask, Spring Boot, React.js, Angular, .NET Framework, Laravel

Data & Systems: PostgreSQL, MySQL, Oracle, Microsoft SQL Server (T-SQL), NoSQL, RabbitMQ, Redis, Microservices, Distributed Systems, Service-Based Architecture, Observability, Performance Engineering, Libsodium, Async Processing, Message Queues

Compliance & Governance: HIPAA, PHI Data Handling, Data Privacy, RBAC, HUD Compliance, Affordable Housing Regulation, Regulated Data Systems, Audit Trails, Zero-Incident Migrations

Process: Git, Linux, Agile/Scrum, Code Review, Technical Mentorship, Cross-functional Delivery, Microsoft Power Apps

EXPERIENCE

Magizhs Tech Solution

May 2025 - Aug 2025

AI/ML Engineer Intern

- Built a production LangGraph agentic system (parse → plan → retrieve → validate → conditional routing) delivering a risk-aware compliance strategy for an external client -- compliant missions route to export, non-compliant ones route to rejection with grounded, cited reasons.
- Parsed natural-language requests from marketing, real estate, and commercial stakeholders into structured plans using quantitative photogrammetry math (GSD, camera footprint, overlap-driven line spacing); implemented RAG over FAA Part 107 corpus with cited retrieval for every compliance check.
- Engineered validation guardrails enforcing altitude limits, battery budgets, and geometric no-fly-zone checks before MAVLink/QGroundControl export; owned the full technical roadmap from model selection through delivery.

George Mason University (Teaching Assistant + Research Associate)

Aug 2024 - May 2026

Graduate Teaching Assistant & Research Associate · College of Engineering and Computing

- Fine-tuned a custom classification model architecture using HuggingFace Transformers and PyTorch; deployed inference API on Quart (async Python web framework); presented model design, training methodology, and evaluation results to faculty.
- Built multi-source ML prediction models (Python, R) on WHO and UK Road Safety datasets -- $R^2 = 0.977$ regression accuracy; applied Isolation Forest for automated anomaly detection across 3 research pipelines.
- Designed Tableau, ggplot2, and Quarto dashboards for faculty leadership; improved research comprehension by 40% across two cohorts.
- Delivered Python programming and database design workshops (SQL, ER modeling, MS Access) to 600+ students across multiple sections of MIS 303; provided one-on-one debugging support and rubric-aligned grading.

Entrata (PropTech SaaS · 100K+ MAU · PHI & HUD compliance · enterprise systems)

Sep 2018 - Aug 2024

Senior Software Engineer · Remote (Lehi, UT)

- Owned the annual HUD income-limit import pipeline for 7 years -- federal source rows expanded into ~180,000 domain records powering eligibility, rent, HAP, and tax-credit calculations for 100K+ housing units; every release included baseline validation, reconciliation, and rollback preparation. Zero compliance violations across the ownership period.

- Built and maintained the RabbitMQ-based async processing system for gross-rent updates across thousands of tenant records -- designed retry logic and idempotency guarantees that eliminated double-processing failures.
- Led the AWS S3 document migration for the Affordable Housing module -- Libsodium-encrypted pipelines, 100% PHI compliance, zero security incidents across the entire resident data migration.
- Optimized 200+ SQL queries (30% performance gain) through execution plan analysis, indexing, partitioning, and stored procedure refactoring across multi-tenant production environments.
- Shipped 20+ full-stack production modules (Python, PHP, PostgreSQL, JavaScript, RabbitMQ) across 120+ Agile sprints; led microservices redesign using Docker (50% scalability improvement, 30% deployment time reduction); eliminated 80% of manual workload through Python/Bash automation pipelines.
- Mentored 5+ junior engineers; conducted 300+ peer code reviews; built internal debugging tooling that reduced investigation time by 40% and kept bug count below 10 for 6+ consecutive months.

Entrata India, Pune

Jul 2017 - Sep 2018

Associate Software Engineer

- Built property management modules (PHP, Python, Java, PostgreSQL) for affordable housing and rental management, including income limits import (1.6M HUD/Tax Credit records across 5 years), document management, and compliance workflows.
- Led cloud migration of client documents from legacy server storage to AWS S3 -- resolved 90% of legacy space constraints; created 56 state-specific housing certification documents with merge fields.
- Redesigned document storage database architecture, migrating hundreds of thousands of records to a modern schema; provided 6 months of post-deployment stabilization.

Entrata India (Db Xento Systems), Pune

Jan 2017 - Jun 2017

Trainee Software Engineer

- Acquired foundational skills in PHP, PostgreSQL, PL/SQL, and Agile practices; contributed to property management and affordable housing data workflows under senior developer guidance.

PROJECTS

AgentProbe: Open-Source MLOps & LLM Evaluation Platform

2025 | github.com/venkatalakshmiparimala/AgentProbe

Python · FastAPI · LangGraph · ChromaDB · CI/CD (GitHub Actions) · Multi-provider LLM routing (Anthropic, OpenAI, Groq, Ollama)

- Open-source LLMOps and evaluation platform: CI-integrated regression gate, LLM-as-judge scoring, hallucination detection, model monitoring, pass@k harness, and multi-provider LLM routing with zero vendor dependency. 5-metric eval engine (correctness, groundedness, hallucination rate, latency, token efficiency); 15-endpoint FastAPI backend; used to evaluate Leasel across 1,240+ production-quality test cases.

Leasel: 8-Agent LangGraph AI System with RAG & HITL

2026 | github.com/venkatalakshmiparimala/Lease_complaince_rag

Python · LangGraph (8-agent) · FastAPI · RAG · OCR · ChromaDB · SQLite · TF-IDF · Streamlit

- Three-surface enterprise AI knowledge platform: Document Intelligence (OCR-fallback ingestion, section-aware chunking, citation correctness, human-review gate); Support Investigation (8-agent pipeline: Triage → Retrieval → Configuration → Historical Incident → Structured Data → Reasoning → Validation → Human Escalation); Tenant Self-Service (policy Q&A; eligibility always routes to staff).
- 93.6% precision, 87.9% F1 across 10 compliance rules on 1,240 residential lease PDFs (5 US states + federal, 1,050 synthetic tenants); pass@1 81.7% / pass@3 88.3% via pass@k harness with multi-trial sampling and LLM-as-judge scoring.

AI on FHIR: HIPAA-Compliant Healthcare AI System

2024 | github.com/venkatalakshmiparimala/AI_on_FHIR

Flask · React · LLMs · FHIR APIs · spaCy · BERT · Embeddings · Docker · HIPAA · RBAC

- HIPAA-compliant NLP platform translating natural-language queries (EN/ES) into FHIR API calls using LLMs, BERT, and embeddings; RBAC access controls enforced at architecture level across structured PHI data. Reduced manual clinical retrieval time ~60%.

CrashClarity: Large-Scale Road Accident Analysis (104K Records)

2025 | Portfolio

Python · R · scikit-learn · Random Forest · Logistic Regression · Feature Engineering · GitHub Pages

- End-to-end ML pipeline on 104K+ UK road accident records -- Random Forest + Logistic Regression with feature engineering across weather, road, and time variables (85%+ accuracy); interactive dashboard deployed on GitHub Pages.

EDUCATION

M.S. Data Analytics Engineering - George Mason University, Virginia — Graduated May 2026 | GPA: 3.87

B.Tech Computer Science & Engineering - JNTU Anantapur — 2013-2017

VOLUNTEER & COMMUNITY

Saving Hands NGO, Pune, India · Volunteer Educator & Support Worker — 2018 - 2024 (concurrent with full-time employment)

- Provided one-on-one academic support to visually impaired children over 6 years -- teaching core subjects, assisting with exam preparation, and adapting instruction to individual learning needs.